A Hybrid Explainable ML Framework for Epileptic Seizure Detection from UCI EEG

HXAI-ML: A Hybrid Explainable ML Framework for Epileptic Seizure Detection from EEG Using Four-Method Consensus Feature Selection and SMOTETomek

1. Abstract

This study benchmarks eleven classifiers for binary seizure detection on the UCI Epileptic Seizure Recognition dataset (11,500 one-second EEG windows of 178 samples). Thirty-seven time-domain, Hjorth, spectral and band-ratio features were appended to the raw samples (215 total) and reduced to 20 by a four-method consensus (Chi², mutual information, RFECV, SFS). SMOTETomek rebalanced the training split from 4:1 to 1:1. On a stratified held-out test set (n = 2,300; 460 seizure), the soft-voting ensemble reached AUC-ROC 0.9985, F1 0.9658 and recall 0.9826; Extra Trees had the highest AUC (0.9986). The top three models are practically indistinguishable at this test size. Because splits are window-level, windows from one recording can fall on both sides of the split, so results should be read as an optimistic upper bound.

2. Problem Formulation

- **Task:** seizure (y = 1) vs non-seizure (y ∈ {2, 3, 4, 5}); `y_binary = (y == 1)`.
- **Why binary:** the clinical question is "is a seizure occurring?"; classes 2–5 are all seizure-free states. Binarising gives one decision threshold and a recall-centred evaluation. The 5-class label is kept only for plots; no multi-class model was trained.
- **Imbalance:** 9,200 : 2,300 = 4:1 (train 6,440 : 1,610; val 920 : 230; test 1,840 : 460).
- **Red flag:** the 4:1 ratio is created by binarising a perfectly balanced 5-class set (2,300 per class). It is not natural prevalence; real seizure prevalence is far lower, so precision and F1 will not transfer to deployment.

3. Dataset

Property	Value
Source	UCI ML Repository (Andrzejak et al., 2001)
Size	11,500 × 178 features + label (ID column dropped); 0 missing values
Sampling rate	173.6 Hz
Window	1 s = 178 samples (23 windows per source recording, per UCI documentation)

y	Meaning (UCI description)
1	Seizure activity
2	EEG from tumour area
3	EEG from healthy brain area
4	Eyes closed
5	Eyes open

Caveats

- **Red flag:** no subject IDs. The dropped first column (e.g. `X21.V1.791`) appears to encode window and source-recording index; if confirmed, recording-level grouping is possible, patient-level is not.
- Windows cut from one recording are highly correlated, so random splitting leaks information.
- The notebook's plot legend names for classes 3–5 differ from the UCI description and should be aligned.

4. Preprocessing Pipeline

Step	Detail
1	Drop ID column; binarise target
2	Stratified split 70/10/20, seed 42 → 8,050 / 1,150 / 2,300
3	StandardScaler fit on train only, applied to val/test (all 215 features)
4	Shift by per-feature train minimum → non-negative input for Chi^2
5	Feature selection fit on train only → 20 features
6	SMOTETomek on the 20-feature train matrix only: 6,440 / 1,610 → 6,439 / 6,439 (12,878)

Tomek cleaning removed one majority sample, so the balancing is effectively SMOTE. Validation and test sets are untouched.

5. Feature Engineering

Category	Feature	Rationale
Time	mean	Baseline offset
Time	std	Amplitude spread
Time	variance	Signal power

Time	skewness	Waveform asymmetry
Time	kurtosis	Spike heaviness
Time	min	Negative extreme
Time	max	Positive extreme
Time	peak_to_peak	Total excursion
Time	rms	Signal energy
Time	mean_abs	Average magnitude
Time	median	Robust centre
Time	iqr	Robust spread
Time	coeff_var	Relative variability
Time	zero_crossing_rate	Frequency proxy
Time	line_length	Cumulative waveform variation
Time	p10 / p25 / p75 / p90	Amplitude distribution shape (4 features)
Hjorth	hjorth_activity	Power (equals variance)
Hjorth	hjorth_mobility	Mean-frequency proxy
Hjorth	hjorth_complexity	Deviation from a pure sine
Spectral	power_delta / theta / alpha / beta / gamma	FFT power in 0.5–4, 4–8, 8–13, 13–30, 30–50 Hz (5 features)
Spectral	rel_delta / theta / alpha / beta / gamma	Band share of total power (5 features)
Spectral	dom_freq	Peak frequency
Spectral	spectral_entropy	Spectral irregularity
Ratio	theta_alpha_ratio	Slow vs alpha activity
Ratio	delta_beta_ratio	Slow vs fast activity
Ratio	gamma_total_ratio	Gamma share of total

Count: 19 time + 3 Hjorth + 12 spectral + 3 ratio = 37.

Red flags: `hjorth_activity` ≡ `variance` and `gamma_total_ratio` ≡ `rel_gamma` (identical formulas), so only 35 features are unique. The function docstring mentions an approximate-entropy proxy that is not implemented. At 1 s and 173.6 Hz, spectral resolution is about 1 Hz, which is coarse for the delta band.

6. Feature Selection

Method	Setup	Output
Chi^2	Scaled, min-shifted train data	Top 50
Mutual information	mutual_info_classif, seed 42	Top 50
RFECV	Random Forest (50 trees, depth 8), 3-fold, AUC, min 20	35 selected
SFS	Forward, Logistic Regression (C = 0.5), 3-fold AUC	20 selected

Consensus rule: documented as "≥ 2 votes", but implemented as the 20 features with the most votes (`vote_df.head(20)`). All 20 have ≥ 3 votes. Ties at 3 votes rely on pandas' default (non-stable) sort, and the notebook does not print how many features reach 3 votes.

Final 20 features

Votes	Features	Missed by
4	std, power_theta, p10, p90, iqr, line_length, power_delta	none
3	min, rms, mean_abs, p75, p25, hjorth_activity, power_alpha, power_beta, power_gamma	SFS
3	zero_crossing_rate, delta_beta_ratio, hjorth_mobility, rel_theta	MI

Why consensus: Chi^2 and MI are univariate and ignore redundancy; RFECV is tied to a tree model; SFS to a linear one. Requiring agreement reduces each method's bias. **Observation:** all 20 are engineered features (no raw sample survived) and most are collinear amplitude/energy statistics, so the set is largely redundant.

7. Modeling

Model	Grid size (n_iter)	Best hyperparameters	Predict, ms*
Logistic Regression	16 (12)	C=50, lbfgs, l2	1.21
Decision Tree	144 (15)	depth 5, min_leaf 10, entropy, max_features None	1.17
Random Forest	320 (5)	500 trees, depth 15, min_leaf 3, max_features 0.3	172.13
Gradient Boosting	576 (5)	200 est., lr 0.05, depth 6, subsample 0.9, min_leaf 5	10.06
HistGradientBoosting	1,920 (20)	max_iter 100, lr 0.2, depth 10, min_leaf 50†	8.58
Extra Trees	256 (20)	300 trees, depth None, min_leaf 1, max_features 0.5	108.10
AdaBoost	20 (12)	200 est., lr 1.0	63.42

KNN	20 (10)	k=11, uniform, euclidean	84.20
SVC	32 (10)	C=100, rbf, gamma scale	59.53
Voting Ensemble	n/a	Soft; LR, RF, GB, HGB, ET (equal weights)	297.31
Stacking Ensemble	n/a	RF, GB, HGB → LR meta-learner (C=1), 3-fold, no passthrough	190.71

*Prediction time for 2,300 windows, single run. Training time per model was not recorded. †L2 and max_bins results were not captured in the outputs.

Tuning used `RandomizedSearchCV` (5-fold stratified, AUC) on the resampled training set. Random Forest and Gradient Boosting sampled only 5 candidates.

8. Evaluation Protocol

- **Metrics:** accuracy, recall, precision, F1, F2, AUC-ROC on the test set; PR curves, confusion matrices and a radar chart in the notebook. PR-AUC is plotted but not tabulated.
- **Why recall/F2:** a missed seizure (false negative) costs more than a false alarm.
- **Selection criterion:** the notebook's summary ranks by AUC-ROC (Extra Trees); the narrative choice of Voting rests on accuracy, F1 and F2. Voting does not lead on AUC, precision or recall.
- **Red flag:** no patient- or recording-wise split. The validation set was scored after the fact, and the winner was picked on test data.

9. Results

Model	Accuracy	Recall	Precision	F1	F2	AUC-ROC
Extra Trees	0.9857	0.9696	0.9591	0.9643	0.9675	0.9986
HistGradientBoosting	0.9857	0.9804	0.9495	0.9647	0.9741	0.9985
Voting Ensemble	0.9861	0.9826	0.9496	0.9658	0.9758	0.9985
AdaBoost	0.9787	0.9848	0.9152	0.9487	0.9700	0.9983
Stacking Ensemble	0.9843	0.9739	0.9492	0.9614	0.9689	0.9983
Gradient Boosting	0.9835	0.9739	0.9451	0.9593	0.9680	0.9981
Random Forest	0.9826	0.9761	0.9393	0.9574	0.9685	0.9980
SVC	0.9835	0.9804	0.9396	0.9596	0.9720	0.9970
Logistic Regression	0.9761	0.9696	0.9158	0.9419	0.9583	0.9954
Decision Tree	0.9722	0.9957	0.8808	0.9347	0.9703	0.9953
K-Nearest Neighbors	0.9765	0.9761	0.9126	0.9433	0.9627	0.9930

Confusion matrices (counts derived from the classification reports; test n = 2,300)

Model	TP	FN	TN	FP
Voting Ensemble	452	8	1,816	24
HistGradientBoosting	451	9	1,816	24
Extra Trees	446	14	1,821	19
AdaBoost	453	7	1,798	42
Decision Tree	458	2	1,778	62

Voting vs Extra Trees differs by six caught seizures and five extra false alarms. The Wilson 95% CI for Voting recall is about 96.6–99.1%, so the top models cannot be separated. Decision Tree maximises recall at the cost of 62 false alarms.

Context: hyperparameter tuning changed AUC little for most models (non-tuned Voting 0.9984, Extra Trees 0.9986); the large exception is Decision Tree (0.9617 → 0.9953). In the balancing ablation, SMOTETomek vs no balancing moved top-model AUC by ≤ 0.0003 and mainly shifted recall against precision.

SHAP (Extra Trees, seizure class), top 5 by mean |SHAP|: std, iqr, rms, line_length, mean_abs. All are amplitude/energy measures; because they are collinear, their relative ranking is unstable. SHAP was not run on the Voting Ensemble.

10. Error Analysis

The notebook does not break errors down by original class (indices are lost in the binary split), so the following is limited to what is measured.

- **False positives:** Voting produces 24 of 1,840 non-seizure windows (1.30%). Which of classes 2–5 they come from is not recorded.
- **False negatives:** 8 of 460 seizure windows (1.74%).
- **Hypothesis to test:** errors concentrate in high-amplitude non-seizure windows and low-amplitude seizure windows, given the amplitude-dominated feature set. Fix: split on an index array and tabulate errors per original label.

11. Reproducibility

Item	Status
Random seed	42 (NumPy and every `random_state`)
Library versions	Not printed; Python 3.12 runtime, scikit-learn ≥ 1.6 implied by warnings. Add `pip freeze`
Hardware	Kaggle notebook (T4 available); all models are scikit-learn CPU, so the GPU is unused
Runtime	Feature extraction 28.0 s (measured); prediction times in §7; other sections not timed (add `%%time`)
Data path	First cell searches `/kaggle/input` only and raises elsewhere

12. Limitations & Threats to Validity

- Single dataset, no external validation.
- Segment-level split may inflate performance through same-recording leakage.
- 1-second windows may miss short events and limit spectral resolution.
- Hyperparameter search, stacking and the learning curve run CV on resampled data, so synthetic samples cross folds and CV scores are inflated.
- Test-set prevalence (20%) is far above clinical prevalence; threshold fixed at 0.5, no calibration.

13. Future Work

- Recording-grouped CV now; patient-wise CV and CHB-MIT cross-validation next.
- Real-time streaming inference.
- Deep-learning baselines (EEGNet, 1D-CNN-LSTM).
- Single-feature (std) baseline and per-class error analysis.

14. References

1. Dataset: https://archive.ics.uci.edu/ml/datasets/Epileptic+Seizure+Recognition
2. Andrzejak et al. (2001). *Phys. Rev. E* 64, 061907.
3. Hjorth (1970). EEG analysis based on time domain properties. *Electroencephalogr. Clin. Neurophysiol.* 29(3).
4. Esteller et al. (2001). Line length: an efficient feature for seizure onset detection. *IEEE EMBS*.
5. Chawla et al. (2002). SMOTE. *JAIR* 16. Batista et al. (2004). *SIGKDD Explorations* 6(1).
6. Guyon et al. (2002). Gene selection for cancer classification using SVMs. *Machine Learning* 46.
7. Lundberg & Lee (2017). A unified approach to interpreting model predictions. *NeurIPS*.
8. Shoeb & Guttag (2010). Application of machine learning to epileptic seizure detection. *ICML* (CHB-MIT).
9. Lawhern et al. (2018). EEGNet. *J. Neural Eng.* 15(5).